
Gains and Collapse in On-Policy Distillation: A Reinforcement Learning Perspective

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Abstract

On-policy distillation (OPD) has become an important approach to language model post-training. However, despite its performance gains, OPD can also collapse into excessively long and repetitive generation, and the mechanism underlying these divergent outcomes remains poorly understood. We explain these outcomes through a reinforcement learning perspective: the teacher implicitly rewards student behaviors, even those it rarely exhibits itself. From this perspective, our experiments show that OPD improves performance without expanding the student’s capabilities. When the implicit reward model is reliable, OPD makes correct responses easier to sample. In contrast, when the preference misaligns with quality, reward hacking happens: the implicit reward model amplifies overlong, repetitive student rollouts, even though it rarely generates such text itself. Guided by this diagnosis, we find that masking unhealthy responses during training and using SFT initialization can each effectively mitigate the collapse. Together, these findings show that OPD amplifies student behaviors favored by the teacher’s implicit feedback, shifting the focus from how well the teacher generates to how reliably it evaluates student rollouts. Our code is available at https://github.com/HancCui/opd_hacking.

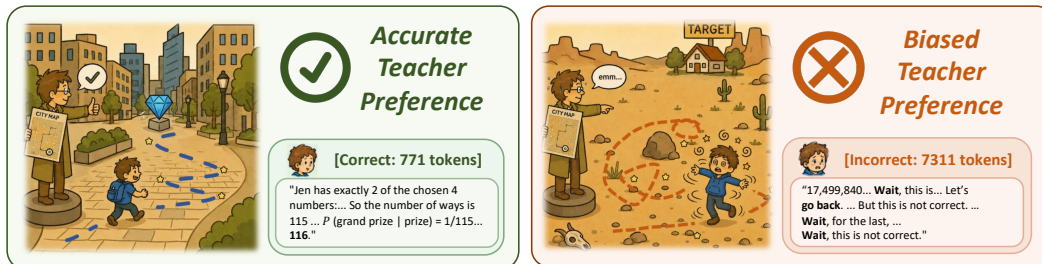


Figure 1: **OPD amplifies what the student already samples.** The teacher scores student-generated trajectories rather than supplying training trajectories of its own; the examples illustrate how its preferences can amplify either coherent or degenerate student behavior.

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1 Introduction

On-policy distillation (OPD) has become an important approach for developing frontier language models and has been adopted in large-scale post-training pipelines [DeepSeek-AI et al., 2026, Kimi Team et al., 2026, Xiao et al., 2026, Yang et al., 2025, Zeng et al., 2026]. At each iteration, the current student samples its own responses, and the teacher supervises it through token-level KL divergence at student-generated prefixes [Agarwal et al., 2024, Gu et al., 2024]. While prior work has demonstrated the potential of OPD to improve student performance, recent studies have reported unstable training and degenerate behaviors, including excessive response length and repetition [Li et al., 2026, Zhu et al., 2026]. It remains unclear what OPD actually learns during training, and why this learning process can sometimes lead to collapse.

In this paper, we view OPD from an RL perspective, in which the teacher implicitly rewards the student’s own responses. We investigate OPD on mathematical tasks using different types of teachers, including RL-trained reasoning models and larger general-purpose models. Our experiments show that **OPD does not directly extend the student’s capability. Instead, it changes the relative probability of responses the student already produces according to the teacher’s preferences.** When these preferences align with response quality, this reweighting can improve sampling efficiency; when they do not, it can lead to reward hacking, with the student increasingly producing responses that receive high implicit rewards despite poor quality. We substantiate this view through three findings.

First, we show that successful OPD reinforces correct responses the student can already generate. We compare the initial and trained students across sampling budgets, combining pass@ k evaluation with additional sampling and manual audits. Although OPD substantially improves single-sample accuracy (pass@1), the gains progressively diminish as the sampling budget increases. At the same time, we find no expansion of the observed solvable set: problems solved after OPD can also be solved by the initial student given sufficient sampling. Together, these results suggest that OPD makes existing capabilities easier to elicit rather than expanding the student model’s problem-solving capabilities.

In contrast, we find that the observed OPD collapse reflects reward hacking: training reinforces poor-quality responses favored by the teacher. We compare a successful OPD run with a reproduction of the collapsed setting reported by Li et al. [2026]. Despite their sharply different outcomes, both runs converge normally and concentrate on responses with low negative log-likelihood under the initial student, suggesting that collapse is not simply a failure of optimization or convergence. In the collapsed run, the student produces increasingly overlong and repetitive responses. At the same time, its response distribution shifts markedly toward responses preferred by the teacher over those the teacher disfavors, indicating that the student is fitting the teacher’s preference signal increasingly well. The student thus learns the teacher’s preferences even as its generation quality deteriorates.

Building on this diagnosis, we mitigate OPD collapse with a standard remedy for reward hacking: limiting the reinforcement of pathological responses. Before training, higher implicit rewards favor correct and concise responses in the successful setting, but overlong and repetitive responses in the collapsed setting, with both types of behavior later amplified during training. Notably, the teacher itself rarely generates such pathological responses, showing that generation quality does not reliably reflect feedback quality on student rollouts. Keeping the teacher fixed, masking rollouts that reach the generation limit mitigates degeneration and improves average evaluation accuracy by 3.08 percentage points, and SFT warmup provides a complementary solution by improving the quality of the student’s initial rollout distribution. Together, these results show that collapse can be mitigated by controlling which student rollouts are available for reinforcement.

Our contributions are summarized as follows:

- We interpret OPD as an RL-like process with the teacher acting as an implicit reward model, linking its performance gains to improved sampling efficiency.
- We identify reward hacking in the observed collapse: the student fits the teacher’s preferences increasingly well even as its generation quality deteriorates.
- We trace the collapse to a biased teacher reward signal. Limiting the reinforcement of this biased signal through masking, or reducing exposure to pathological rollouts through SFT warmup, mitigates collapse.

Table 1: OPD settings used in our analysis. Each setting is named after its teacher.

Teacher	Student initialization	Training data
JustRL-1.5B	DeepSeek-R1-Distill-Qwen-1.5B	DeepMath-103K (≥ 6)
DeepScaleR-1.5B-Preview	DeepSeek-R1-Distill-Qwen-1.5B	DeepMath-103K (≥ 6)
Qwen3-4B	Qwen3-1.7B-Base	OpenThoughts3

2 On-policy distillation as implicit reward optimization

We formulate the sampled-token reverse-KL variant of on-policy distillation (OPD) studied in this paper as policy optimization with a teacher-derived reward. This formulation makes explicit what the teacher scores, how those scores guide student updates, and why their reliability must be examined on student-generated trajectories.

2.1 On-policy teacher supervision

Let $q \sim \mathcal{D}$ be a training question with prompt x_q . We denote the fixed teacher by T , the initial student by S_{INIT} , the trainable student by S_θ , and the final trained student by S_{OPD} . At each rollout iteration, the current student policy $\bar{S} = S_{\bar{\theta}}$ generates a response $y = (y_1, \dots, y_L)$. The teacher evaluates tokens conditioned on the resulting prefixes $h_t = (x_q, y_{<t})$.

Let $d^{\bar{S}}$ denote the prefix distribution induced by the rollout policy. The conditional reverse-KL objective is

$$\mathcal{L}_T(\theta; \bar{S}) = \mathbb{E}_{h \sim d^{\bar{S}}} \left[D_{\text{KL}} \left(\pi_{S_\theta}(\cdot | h) \parallel \pi_T^{(\tau_T)}(\cdot | h) \right) \right], \quad (1)$$

where τ_T is the teacher scoring temperature. We hold the rollout-prefix distribution fixed during each optimization step and refresh it through subsequent student rollouts. The student thus determines where supervision is evaluated, while the teacher supplies the target token probabilities at those states.

2.2 The teacher as an implicit reward model

Reward interpretation. By expanding the KL divergence in Equation (1), we can recast the objective into a maximum-entropy policy objective over sampled prefixes in Equation 2:

$$\mathcal{L}_T(\theta; \bar{S}) = -\mathbb{E}_{h \sim d^{\bar{S}}} \left[\mathbb{E}_{a \sim \pi_{S_\theta}(\cdot | h)} [r_T(h, a)] + \mathcal{H}(\pi_{S_\theta}(\cdot | h)) \right], \quad (2)$$

where $r_T(h, a) = \log \pi_T^{(\tau_T)}(a | h)$ and H denotes entropy. Minimizing the conditional reverse KL is therefore equivalent to maximizing the expected teacher reward with entropy regularization on the sampled prefix distribution. This is the token-level reward interpretation underlying policy-gradient implementations of OPD [Lu and Thinking Machines Lab, 2025].

Policy-gradient update. The reward interpretation also yields the sampled-token update. For each rollout token, define

$$A_t^T = r_T(h_t, y_t) - \log \pi_{\bar{S}}(y_t | h_t). \quad (3)$$

Writing $A^T(h, a) = r_T(h, a) - \log \pi_{\bar{S}}(a | h)$, the unclipped estimator at a fixed prefix h satisfies

$$\begin{aligned} & \mathbb{E}_{a \sim \pi_{\bar{S}}(\cdot | h)} [A^T(h, a) \nabla_\theta \log \pi_{S_\theta}(a | h)] \Big|_{\theta=\bar{\theta}} \\ &= -\nabla_\theta D_{\text{KL}} \left(\pi_{S_\theta}(\cdot | h) \parallel \pi_T^{(\tau_T)}(\cdot | h) \right) \Big|_{\theta=\bar{\theta}}. \end{aligned} \quad (4)$$

In practice, we treat A_t^T as a stop-gradient advantage in a PPO-style clipped surrogate and add reference-KL regularization toward S_{INIT} . The teacher score r_T and the update weight A_t^T are distinct: the latter includes the student log-probability term arising from entropy regularization. These updates use immediate token-level signals on student-generated prefixes. When most advantages are negative, the main effect is to reduce token probabilities by different amounts. More negative advantages lead to stronger pressure to reduce the probability, while less negative advantages lead to weaker pressure.

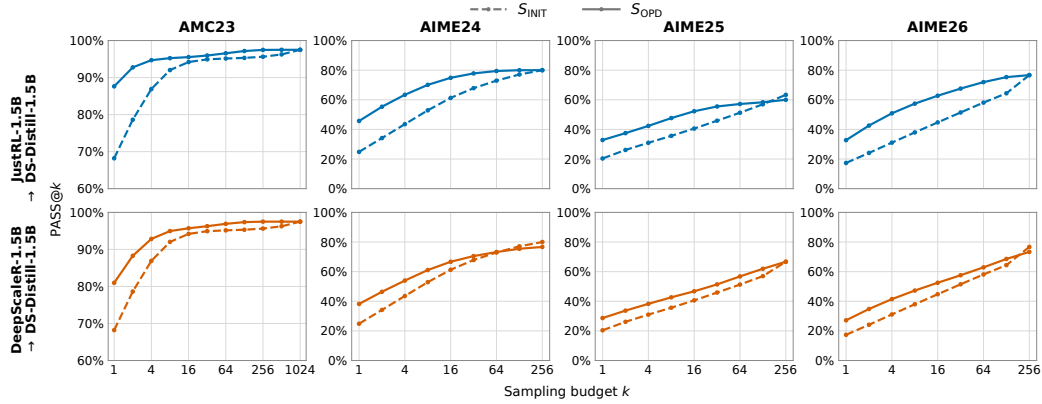


Figure 2: **PASS@ k curves of S_{INIT} and S_{OPD} .** Across AMC23 and AIME24–26, OPD gains are largest at small k and shrink as the sampling budget grows. AIME curves extend to $k = 256$, while AMC23 extends to $k = 1024$.

2.3 Experimental settings

We study three teacher–student settings, summarized in Table 1. The JustRL and DeepScaleR settings share the same student initialization and training data. Their teachers, JustRL-1.5B [He et al., 2025] and DeepScaleR-1.5B-Preview [Luo et al., 2025], were obtained through RL from DeepSeek-R1-Distill-Qwen-1.5B [Guo et al., 2025], which also initializes our student. Both settings use the level ≥ 6 subset of DeepMath-103K and a 16,384-token rollout limit. The Qwen3-4B setting follows Li et al. [2026], using Qwen3-4B to supervise Qwen3-1.7B-Base on OpenThoughts3 prompts [Guha et al., 2025] with an 8,192-token rollout limit. The teacher remains fixed in all settings. We train for 100 rollout iterations using the Slime training framework [Zhu et al., 2025], with a reference-KL coefficient of 0.001. We evaluate mathematical reasoning on AIME24–26, reporting mean@4 for routine evaluation and using 256 responses per problem for the coverage and sampling efficiency analyses. Full training and evaluation protocols are provided in Appendix A.1.

3 Does OPD Expand Student’s Capability Boundary?

The implicit-reward perspective suggests that successful OPD can improve performance by reinforcing correct student responses. We examine this successful case: does teacher feedback make problems already solvable by the initial student easier to solve? We compare the initial student S_{INIT} with the trained student S_{OPD} using $\text{pass}@k$ across increasing sampling budgets, audited problem coverage, and per-problem success rates.

Gains diminish as the sampling budget grows. We compare the initial student S_{INIT} with the trained student S_{OPD} under both teachers, JustRL-1.5B and DeepScaleR-1.5B-Preview, on AIME24–26 and AMC23. We compute $\text{PASS}@k$ from each response pool for $k \in \{1, 2, 4, 8, 16, 32, 64, 128, 256\}$ on AIME and additionally $k \in \{512, 1024\}$ on AMC23, averaged across all test cases.

As shown in Figure 2, we observe a consistent trend under both teachers. When k is small (e.g., $k = 1$, equivalent to average-case accuracy), S_{OPD} clearly outperforms S_{INIT} . However, as k increases, S_{INIT} steadily catches up. At the maximum available k , $\text{PASS}@k$ equals the observed coverage: the fraction of problems solved at least once in the response pool. This suggests that OPD improves accuracy under a small sampling budget, but does not improve the observed coverage at higher k .

No Observed Expansion in Problem Coverage After OPD. To distinguish new coverage from finite-sample misses, we first identify the problems solved by S_{OPD} but not by S_{INIT} within the original 256-response pools of AIME24–26. For each of these problems, we draw 768 additional responses from S_{INIT} , bringing its pool to 1024 responses. Manual review checks derivation validity and corrects verifier misses in the existing outputs.

After additional sampling and manual review, no problem remains solved only by S_{OPD} : candidate cases are resolved either by finding a valid S_{INIT} solution or by excluding an invalid S_{OPD} derivation. The audited S_{INIT} set contains 73 solvable problems, against which S_{OPD} covers 65 under JustRL-1.5B and 63 under DeepScaleR-1.5B-Preview. These counts are larger than the $k = 256$ endpoints of Figure 2, which are computed before the audit. Detailed coverage analyses are provided in Appendix B. These indicate that OPD does not induce an observed expansion of the solvable set.

Already-solvable problems become easier to solve.

To identify the source of OPD’s small- k gains, we compare per-problem success rates between S_{INIT} and S_{OPD} . For each problem q , let $\hat{p}_m(q)$ denote the fraction of correct responses among the 256 samples from model m .

As shown in Figure 4, the problem-level distribution is strongly shifted above the diagonal under both teachers. OPD increases the per-problem success rate for 90.6% of the problems already solvable by S_{INIT} with JustRL-1.5B as the teacher, and for 84.4% with DeepScaleR-1.5B-Preview. Averaged over these problems, the increase in mean@256 is 22.6% and 14.7%, respectively.

Problems unimproved by OPD are those that S_{INIT} could hardly solve, and no high-performing problem drops to near-zero accuracy (lower-right corner). Some even jump from low to high success rates (upper-left corner), showing that OPD can make problems the initial student rarely solves much easier to solve. Thus, OPD primarily eases sampling of correct answers on already-solvable problems, which accounts for the small- k performance gain.

Summary. In the two successful settings studied here, OPD improves sampling efficiency on problems already solvable by the initial student. Its positive accuracy gains disappear at the largest sampling budgets tested, and no validated OPD-only problem remains after the targeted AIME coverage audit. These findings are consistent with the teacher acting as an implicit reward model whose feedback reinforces correct student behavior, making successful responses easier to sample.

4 Reward Hacking in OPD: Convergence to a Biased Reward Signal

The preceding analysis shows that OPD improves performance without expanding the set of problems the student can solve. Yet OPD training does not always succeed: some teacher-student pairs produce collapsed behaviors instead. From a diagnostic perspective, we then examine whether the collapse constitutes reward hacking: optimization succeeding at the wrong target. We first verify that the optimization converges normally, then characterize the behavioral shifts that emerge during training.

4.1 Collapsed OPD Training Still Converges Normally

Our analysis starts by comparing a reported failure case of OPD with a successful one. Prior work [Li et al., 2026] reports that OPD can fail when the teacher and student distributions are substantially mismatched, attributing the resulting degradation to this large distribution gap. We therefore compare the full training process of the two settings: the **JustRL setting** (JustRL-1.5B $\rightarrow$ DS-Distill-1.5B) where training improves the student, and the **Qwen3-4B setting** (Qwen3-4B $\rightarrow$ Qwen3-1.7B-Base,

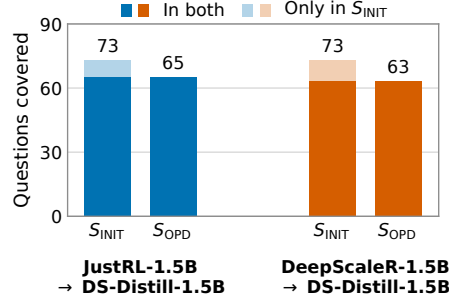


Figure 3: **Change in observed coverage from S_{INIT} to S_{OPD} .** No problem is solvable by S_{OPD} but not by S_{INIT} in our observation. Counts come from the audited pools with conditional extension to 1,024 responses.

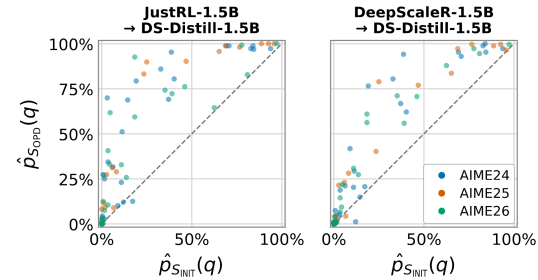


Figure 4: **Per-problem mean@256 of S_{INIT} (x-axis) and S_{OPD} (y-axis).** OPD raises success rates on problems already solvable by S_{INIT} . The dashed line marks equal score.

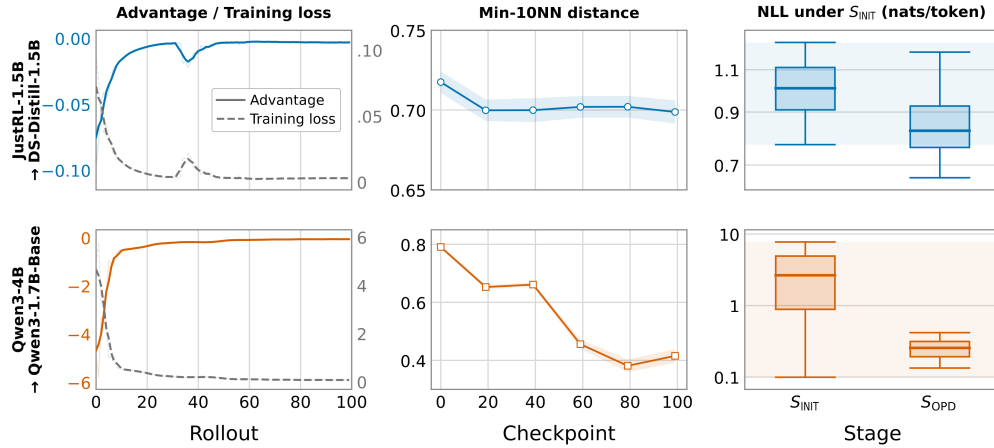


Figure 5: **Training process of the JustRL (top) and Qwen3-4B (bottom) settings.** Optimization converges normally in both settings, and the trained student’s responses remain highly likely under S_{INIT} . Columns: the advantage and training loss; the Min-10NN distance over six checkpoints; the NLL under S_{INIT} of responses from S_{INIT} and S_{OPD} .

without warm-up) where training collapses. The loss and advantage in the Qwen3-4B setting converge as smoothly as those in the JustRL setting, with no qualitative difference in the shape of the curves.

We measure response concentration using the Min-10NN distance [Zhang et al., 2026], which averages the ten smallest nearest-neighbor distances among responses to the same problem; lower values indicate greater similarity (described in detail in Appendix C). The distance keeps decreasing during training in both settings: mildly in the JustRL setting, and much more sharply in the Qwen3-4B one. Both students shrink toward a narrower space of behaviors, a phenomenon also reported in other work on on-policy RLVR post-training [Yue et al., 2025, Zhao et al., 2025, Dang et al., 2025].

If training exploits the reward rather than learning from it, the selected behaviors should already be accessible to S_{INIT} rather than genuinely novel. To examine this, we compute the NLL under S_{INIT} for responses generated by the trained student and compare it with the NLL of S_{INIT} ’s own responses to the same problems. Figure 5 summarizes these diagnostics; the right panel reports this NLL comparison. In both settings, the trained student’s responses lie in the low NLL region of S_{INIT} , which indicates that they are already familiar to S_{INIT} .

4.2 The Student Faithfully Fits the Reward Signal, Even When the Teacher Is Hacked

If the collapse constitutes reward hacking, the trained student should have successfully internalized the teacher’s preferences, but those preferences should be misaligned with solution quality.

Endpoint behaviors are pathological. Table 2 compares the free generations of the teacher, S_{INIT} , and S_{OPD} across both settings. In the JustRL setting, the trained student’s behaviors closely match the teacher’s: accuracy improves by 16.2 points, and no severe repetition is detected. In contrast, the Qwen3-4B setting exhibits a dramatically different outcome. Although accuracy improves modestly (+5.2 points), nearly all responses hit the 8k generation limit (99.4%), and 38% exhibit severe repetition—far worse than either the teacher (2.1% truncation, 0% repetition) or S_{INIT} itself. The trained student has not learned to reason like the teacher; instead, it has concentrated probability mass on a narrow subset of pathological behaviors.

Table 2: **Endpoint generation characteristics on AIME24–26.** Model pairs denote teacher → student. Bold values mark length-limit saturation and severe repetition.

Model	Acc. (%)	Len. (k)	Trunc. (%)	Rep. (%)
JustRL-1.5B → DS-Distill-1.5B				
Teacher	39.3	9.30	13.8	0.0
S_{INIT}	20.9	11.19	12.5	0.0
S_{OPD}	37.1 (+16.2)	9.72	16.5	0.0
Qwen3-4B → Qwen3-1.7B-Base				
Teacher	19.6	3.09	2.1	0.0
S_{INIT}	0.2	1.18	4.1	4.2
S_{OPD}	5.4 (+5.2)	8.16	99.4	38.0

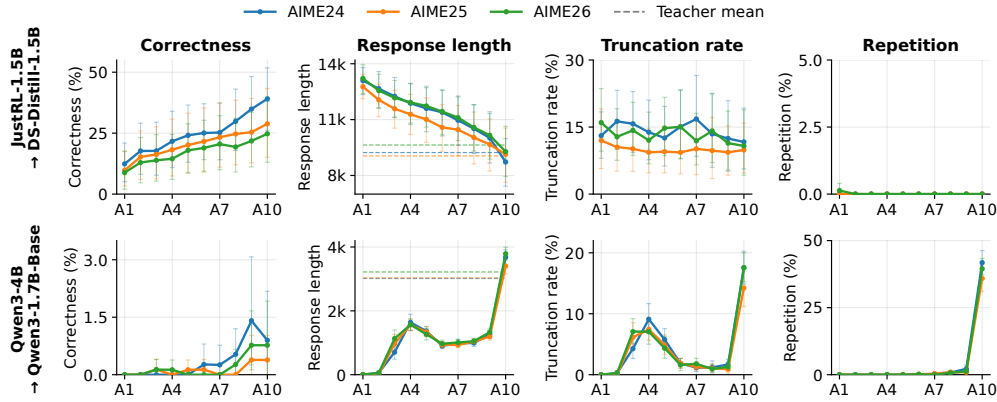


Figure 7: Teacher advantage over S_{INIT} responses from A1 (lowest advantage) to A10 (highest advantage). The JustRL teacher (top) assigns higher advantage to correct and shorter responses; the Qwen3-4B teacher (bottom) assigns higher advantage to overlong, repetitive ones. Dashed lines mark the teacher’s own mean length.

The trained student discriminates teacher-preferred responses. We measure how the trained student scores a fixed set of S_{INIT} responses. We compare the 10% of responses the teacher prefers most with the 10% it prefers least, as ranked by teacher advantage. For each response, $\Delta\text{NLL} = \text{NLL}_{S_i} - \text{NLL}_{S_0}$ measures the change in the student’s NLL during training. The *selection gap* Γ is the mean ΔNLL of the least-preferred 10% minus that of the most-preferred 10%. A positive Γ indicates that training favors the teacher’s most-preferred responses over its least-preferred ones.

Figure 6 shows whether this degradation reflects learned teacher preference rather than random drift. Both students increase their discrimination of teacher-preferred responses: JustRL’s gap grows modestly from near zero to 0.07, while the Qwen3-4B setting’s gap increases dramatically to approximately 26. Combined with the endpoint generation table, this reveals that the student in the Qwen3-4B setting has *overfit* to the subset of S_{INIT} responses that receive high teacher advantage but exhibit pathological behaviors.

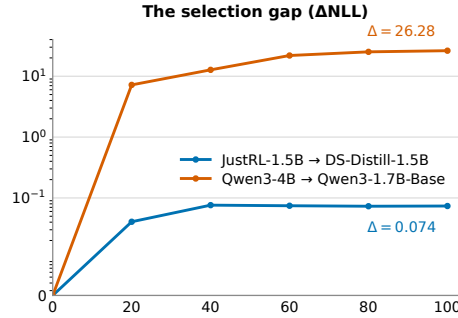


Figure 6: Selection gap before and after training. Both students learn to discriminate teacher-preferred responses, with the Qwen3-4B setting showing overfitting.

5 Diagnosing and Mitigating Reward Hacking

To address reward hacking, prior work identifies which training samples exploit the reward model, then removes or reweights them during optimization [Liu et al., 2025a, Beigi et al., 2026, Zhang et al., 2025]. Following this strategy, we first diagnose which S_{INIT} responses the teacher assigns high advantage to, then test whether masking pathological responses or reinitializing from a cleaner distribution can prevent the collapse.

5.1 The Teacher’s Reward Signal Can Be Misaligned with Response Quality

To characterize the responses favored by the teacher before training, we rank all sampled S_{INIT} responses by their mean teacher advantage and divide them into ten equally sized groups, A1–A10. Each group contains 10% of the responses, with A1 containing the lowest-advantage responses and A10 the highest. Figure 7 reports correctness, response length, generation-cap hit rate, and severe-repetition rate across advantage deciles. In the JustRL setting, teacher advantage is aligned with response quality: correctness increases toward A10, response length decreases toward the

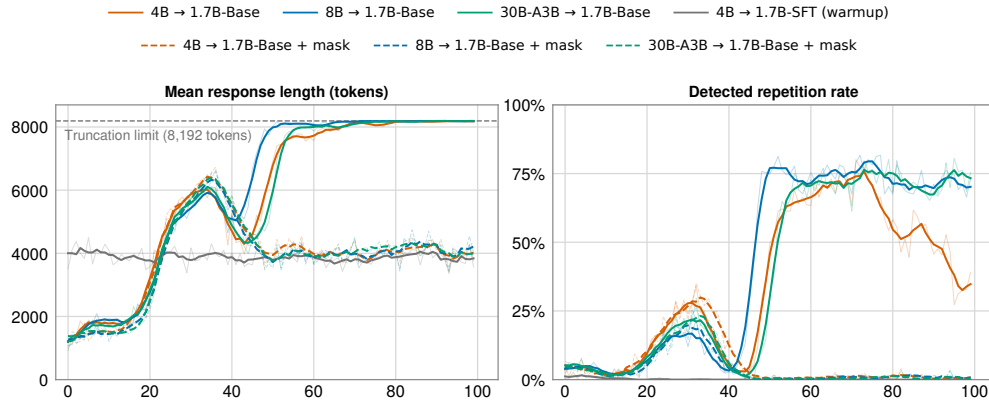


Figure 8: **Student-rollout characteristics during training.** All the masking settings recover from the collapse, whereas Qwen3-4B-warmup avoids it from the beginning.

teacher’s own average, and no severe repetition is detected. The Qwen3-4B setting shows the opposite pattern. The highest-advantage responses are not the most reliable ones: their correctness remains close to zero, whereas their length, truncation rate, and repetition rate increase substantially across advantage deciles. Although these pathological responses are rare under S_{INIT} , the teacher assigns them high advantage, causing OPD to amplify them during training. We also find similar failure pattern when the teacher is replaced as Qwen3-8B and Qwen3-30B-A3B. Appendix E presents three high-advantage S_{INIT} responses that exhibit excessive length and mechanical repetition.

5.2 Candidate-Pool Interventions Mitigate Reward Hacking

The preference bias discussed previously is already visible in responses sampled from S_{INIT} , before any OPD update takes place. This suggests that the collapse may depend on which responses are supplied to the teacher for supervision. To test this hypothesis, we intervene on the candidate pool in the Qwen3-4B setting while keeping the teacher, training data, and optimization recipe fixed. We consider two interventions: (1) The **Qwen3-4B-masked setting**: masking the loss on responses that reach the generation length limit. (2) The **Qwen3-4B-warmup setting**: changing only the student initialization, from Qwen3-1.7B-Base to an SFT-initialized student released by Li et al. [2026]. Appendix F provides the training details for both interventions.

Masking recovers from collapse, while warmup avoids it. Figure 8 shows how response length, truncation, and repetition evolve over the course of training. The original Qwen3-4B setting eventually enters a persistent collapsed state, where responses become increasingly long, truncated, and repetitive. For the Qwen3-4B-masked setting, the student model exhibits few abnormal behaviors in the early training phase, so its learning trajectory closely aligns with that of the original Qwen3-4B. As the undesirable behaviors accumulate, the effect of the masking intervention becomes clear: the Qwen3-4B-masked setting recovers, while the original setting collapses. On the other hand, the warmup setting never develops this failure mode. Warmup has been adopted as a remedy for OPD

Table 3: **Masking improves average accuracy across teachers.** Accuracy (%) and the four-benchmark mean; teachers are Qwen3 models, and Base/SFT denote Qwen3-1.7B initializations.

Setting	AMC	AIME				Avg.
		23	24	25	26	
Base (pre-OPD)	2.41	1.67	0.00	1.67	1.64	
4B → Base	29.82	3.33	3.33	4.17	10.16	
+ mask	34.64	8.33	5.83	4.17	13.24	
8B → Base	28.31	5.00	3.33	5.00	10.41	
+ mask	32.83	5.00	2.50	3.33	10.92	
30B-A3B → Base	28.61	6.67	2.50	4.17	10.49	
+ mask	33.13	10.00	4.17	5.00	13.07	
4B → SFT [†] (warmup)	38.86	9.17	10.00	7.50	16.38	

[†] Warmup uses the Qwen3-1.7B-SFT checkpoint released by Li et al. [2026].

collapse [Li et al., 2026]. The SFT-initialized student rarely produces the pathological responses favored by the teacher, leaving fewer such responses for OPD to reinforce. Together, these results suggest that the collapse is not irreversible; it depends on whether abnormal responses continue to receive reinforcement during training.

Both interventions improve final accuracy. Table 3 extends the accuracy comparison to Qwen3-8B and Qwen3-30B-A3B teachers. Masking improves the average accuracy over AMC23 and AIME24–26 by 3.08, 0.51, and 2.58 percentage points for the 4B, 8B, and 30B-A3B teachers, respectively. These gains are not uniform across benchmarks: with the 8B teacher, AIME24 accuracy is unchanged while AIME25 and AIME26 accuracy decrease from 3.33% to 2.50% and from 5.00% to 3.33%, respectively. The 4B warmup setting starts from a stronger SFT student and achieves the best final average accuracy (16.38%), exceeding the corresponding Base-initialized OPD setting by 6.22 points. In the 4B masking setting, accuracy improves even though only approximately 10% of the sampled responses retain nonzero loss after abnormal responses are masked.

6 Related Work

Distillation paradigms and capability transfer. Knowledge distillation matches teacher distributions [Hinton et al., 2015] or trains on teacher sequences [Kim and Rush, 2016]; for reasoning models, the supervision is teacher rationales and trajectories [Ho et al., 2023, Hsieh et al., 2023, Mukherjee et al., 2023, Mitra et al., 2023, Guo et al., 2025], which extend coverage beyond the base model [Yue et al., 2025]. OPD instead supervises student-generated trajectories at student-visited states [Gu et al., 2024, Agarwal et al., 2024], with extensions to self-distillation [Zhao et al., 2026, Ye et al., 2026, He et al., 2026b], representation-level supervision [Yang et al., 2026a], multiple teachers [Ma et al., 2026, Yang et al., 2026b, Kimi Team et al., 2026], and cross-tokenizer transfer [He et al., 2026a, Wang et al., 2026a]. These broaden teacher supervision but retain the reliance on student rollouts, over which our runs mainly redistribute probability rather than expanding the student’s observed solvable set.

Failure modes and stabilization of OPD. OPD failures arise from mismatches among student rollouts, teacher supervision, and token-level objectives. Li et al. [2026] show that a stronger teacher need not help under reasoning-pattern mismatch; Fu et al. [2026] and Zhu et al. [2026] identify unreliable guidance on student prefixes and bias from approximate reverse-KL updates. Remedies include off-policy warm starts, data selection, stop-gradient objectives, trust regions [Xing et al., 2026], and prefix correction [Jiang et al., 2026]. Rollout-signal feedback creates another failure class: Wang et al. [2026b] identify signal-correctness mismatch and length exploitation, while Luo et al. [2026] link length inflation to truncation and repetition collapse, motivating clipping, reference regularization, rollout mixing, and length control [He et al., 2026a, Heo et al., 2026]. We instead view OPD as selecting among student-accessible behaviors by teacher preference, and analyze how biases in that preference affect training.

Rollout pathologies in on-policy post-training. On-policy optimization trains only on trajectories sampled from the current model and is therefore exposed to the pathologies of its own rollout distribution. For RLVR, pass@k and likelihood analyses show that training mainly redistributes probability over solution paths already accessible to the base model [Yue et al., 2025], and related studies find that reinforcement learning can amplify behaviors acquired during pretraining [Zhao et al., 2025] and reduce the diversity of reasoning trajectories [Dang et al., 2025]. Rollouts can further carry degenerate patterns such as malformed formats, excessive length, truncation, and self-reinforcing repetition. Repetition can be reinforced by token co-occurrence effects [Yan et al., 2024], while RLVR systems counter rollout pathologies with format rewards [Guo et al., 2025], length-debiased advantage estimation [Liu et al., 2025b], and dynamic sampling with overlong filtering [Yu et al., 2025]. OPD inherits this dependence on student rollouts but evaluates them through a teacher-derived signal; we show that degenerate student behaviors can bias the induced preference itself and thereby become preferentially retained during training.

7 Conclusion

We interpret on-policy distillation as an RL-like process in which the teacher acts as an implicit reward model for the student’s own responses. Across our experiments on competition mathematics, OPD improves sampling efficiency without expanding the observed solvable set. Its gains primarily

come from making correct responses already accessible to the initial student easier to sample. This view also helps explain why OPD can collapse despite normal convergence. In the collapsed setting, the student increasingly follows the teacher’s preferences while producing longer and more repetitive responses. These responses receive high teacher advantage even before training, although the teacher rarely generates them itself. Masking the loss on truncated responses and initializing the student from an SFT checkpoint mitigate this failure while keeping the teacher fixed.

Limitations

Computational constraints limit our experiments to small models and mathematical reasoning tasks, where prior work has reported clear cases of OPD success and collapse. Whether our findings extend to larger models or other domains remains to be studied. Our interventions are designed specifically to address the overlong and repetitive generations observed in our experiments. We use them mainly to test our explanation of OPD collapse, rather than propose them as general solutions for OPD training. Finally, our coverage results are based on a finite number of samples. Although we use additional sampling and manual inspection to check apparently new solutions, we cannot rule out that larger sampling budgets or different decoding settings would uncover additional correct responses from the initial model.

AI Use Statement

We used generative AI tools to polish the manuscript, assist with writing portions of the experimental code, and design the cartoon characters in the main figure. The authors take responsibility for the final manuscript, experimental implementation, and figures.

Ethics Statement

This work studies the reasoning behavior and training dynamics of language models using publicly available models and mathematical reasoning datasets. We do not identify ethical concerns specific to the experiments presented here beyond those associated with the underlying models and datasets.

Reproducibility Statement

All models and datasets used in this work are publicly available, either through open releases or accompanying prior publications. The appendix describes the training configurations, evaluation protocols, and analysis procedures. We plan to release all evaluation and analysis scripts to facilitate reproduction of our results.

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A Experimental Details

A.1 Training Settings

We conduct three main experiments to study these phenomena. The JustRL and DeepScaleR settings share the same student initialization and training data, while the Qwen3-4B setting uses a base Qwen3 student and OpenThoughts3 prompts. The differences between the settings are summarized in Table 4.

Table 4: **Training settings used in the main experiments.** The teacher is fixed throughout OPD training.

Setting	Teacher	Student initialization	Training data	Length limit	LR
JustRL	JustRL-1.5B	DeepSeek-R1-Distill-Qwen-1.5B	DeepMath-103K, level ≥ 6	16,384	10^{-5}
DeepScaleR	DeepScaleR-1.5B-Preview	DeepSeek-R1-Distill-Qwen-1.5B	DeepMath-103K, level ≥ 6	16,384	10^{-5}
Qwen3-4B	Qwen3-4B	Qwen3-1.7B-Base	OpenThoughts3 OPD 30K	8,192	10^{-5}

Each setting runs for 100 rollout iterations. At each iteration, the student samples 8 responses for each of 32 prompts at temperature 1.0 and top- p 1.0, producing 256 responses and two optimizer updates with global batch size 128. We use Adam with $\beta_1 = 0.9$, $\beta_2 = 0.98$, weight decay 0.1, no learning-rate warmup, PPO clipping $\epsilon = 0.2$, and an approximate reference-KL coefficient of 0.001 toward the fixed initialization S_{INIT} .

The loss mask covers response tokens only. Each sample is first averaged over its unmasked response tokens, and the resulting values are then averaged over the global batch, so that long and short responses contribute equally. All runs use eight NVIDIA H100 GPUs. Checkpoints are named by rollout count: the initialization S_{INIT} together with iterations 20, 40, 60, 80, and 100, where iteration 100 gives S_{OPD} .

A.2 DeepScaleR Teacher-Temperature Pilot

We ran two small temperature studies before fixing the DeepScaleR setting. First, Table 5 evaluates the frozen DeepScaleR-1.5B-Preview teacher itself. For every temperature, we draw one response per problem at top- p 1.0 with an 8,192-token limit and score it with the verl verifier. The macro average weights AMC23, AIME24, and AIME25 equally despite their different problem counts.

Table 5: **Standalone teacher accuracy depends on the generation-sampling temperature.** Accuracy is in percent; Macro gives equal weight to the three benchmarks, and bold marks each column maximum.

Temperature	AMC23	AIME24	AIME25	Macro
0.1	66.27	23.33	30.00	39.87
0.5	67.47	30.00	26.67	41.38
0.75	65.06	36.67	23.33	41.69
1	56.63	20.00	10.00	28.88
1.5	0.00	0.00	0.00	0.00

The training pilot in Figure 9 instead varies the temperature applied to the teacher logits when scoring student rollouts. Each temperature corresponds to one training run. At each displayed checkpoint, we evaluate the student with four responses per problem sampled at temperature 1.0, top- p 1.0, and a 16,384-token limit; we average within each problem and then give the three benchmarks equal weight. The $T = 1.5$ run has observations only through iteration 20.

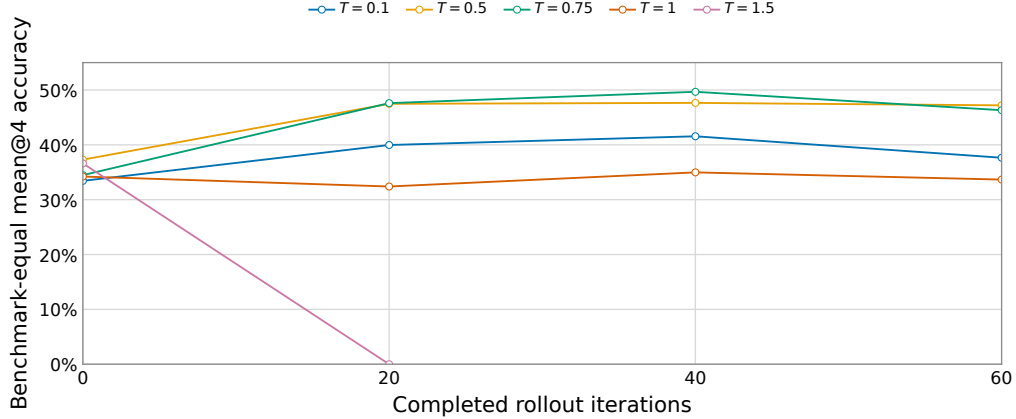


Figure 9: **Intermediate teacher-scoring temperatures give the strongest single-run student trajectories in the pilot.** Benchmark-equal student mean@4 accuracy is plotted against completed rollout iterations for the five teacher-scoring temperatures.

These pilots support only a working choice, not an optimum. In the standalone teacher study, $T = 0.75$ has macro accuracy 41.69%, just 0.31 percentage points above $T = 0.5$; weighting all 143 problems equally reverses the ordering (50.35% versus 51.05%). The teacher study uses one sample per problem, and every training curve is a single run with unequal coverage at $T = 1.5$. Together with the strongest observed training point at $T = 0.75$, these results motivated adopting 0.75 as the working temperature, but do not establish that it is optimal.

A.3 Evaluation Protocol

We use two evaluation protocols, and they differ only in how many responses are drawn per problem. Both sample at top- p 1.0, use the same generation length limit as training, and serialize prompts with the template described in Appendix A.4. The sampling temperature follows the model being evaluated rather than the protocol. In the JustRL setting and the DeepScaleR setting, S_{INIT} and S_{OPD} are both sampled at $T = 1.0$, matching the rollout temperature used in training.

Routine evaluation. Endpoint and per-checkpoint accuracies are reported as mean@4: we draw four responses per problem, and calculate average accuracy among all the responses. This protocol is used for the accuracy columns in Section 5.2 and for the accuracy traces over training.

Large-pool evaluation. The coverage and PASS@ k analyses in Section 3 require a much larger budget. For each model and each of the 90 AIME24–26 problems we draw $n = 256$ responses. This pool supports PASS@ k for $k \in \{1, 2, 4, 8, 16, 32, 64, 128, 256\}$ and the per-problem success rate $\hat{p}_m(q)$, defined as the fraction of verifier-correct responses among the 256 samples.

Conditional extension for the coverage audit. For the audit in Appendix B we extend sampling only where it can change the conclusion. For every problem covered by S_{OPD} but not by S_{INIT} at $n = 256$, we draw a further 768 responses from S_{INIT} , giving a conditional pool of 1,024 responses for that problem. The prompt, temperature, top- p , and length limit are unchanged; only the number of samples increases.

The audit protocol is deliberately one-sided. Both halves of the audit act in a single direction, and we state this explicitly rather than leave it to be inferred. The additional sampling is spent only on problems that S_{OPD} covers and S_{INIT} does not; the reverse case—covered by S_{INIT} but not by S_{OPD} —never receives a larger S_{OPD} pool. Manual review is applied to the same candidate list, where it reinstates verifier misses among S_{INIT} responses and rejects invalid S_{OPD} derivations. Relative to the unaudited 256-response pools, the procedure can therefore only raise S_{INIT} coverage and only lower S_{OPD} coverage, which is the direction that favors the conclusion of Section 3.

We chose this design because a symmetric audit at equal budget would spend most of it on problems whose status is not in question, and because the claim under test is one-sided: a single surviving S_{OPD} -only problem would refute it, whereas additional S_{INIT} -only problems would not change it. The cost is that the audited partition in Table 6 is not a symmetric measurement of the two models. It should be read as an upper bound on S_{INIT} coverage and a lower bound on S_{OPD} coverage under this protocol, so the empty S_{OPD} -only column reports that no candidate survived a search directed against it, not that the two supports were compared on equal terms. The audited S_{INIT} and S_{OPD} totals are therefore not on the same footing, and their difference overstates the true gap.

A second consequence is that the candidate list differs between the two teacher settings, so the two rows of Table 6 audit the same S_{INIT} checkpoint under different budgets. We resolve this by reporting a single S_{INIT} set: the union of the problems it is found to solve under either audit. This is sound because S_{INIT} is one checkpoint drawing on one shared pool of 256 responses per problem, so a solution recovered in one audit is evidence about the checkpoint and not about the teacher it was audited alongside. No counterpart exists on the S_{OPD} side, where the two rows are different models. Pooling in this way gives S_{INIT} the combined budget of both audits and so widens the asymmetry above; it cannot create an S_{OPD} -only problem, since enlarging S_{INIT} can only remove entries from that column.

Estimating PASS@k. Reporting PASS@k from a single group of k responses is high variance. For a problem with c correct responses among n , we instead use the standard unbiased estimator of the probability that a group of k responses drawn without replacement from the pool contains at least one correct response [Chen et al., 2021],

$$\text{PASS@}k = \mathbb{E}_q \left[1 - \frac{\binom{n-c_q}{k}}{\binom{n}{k}} \right], \quad (5)$$

evaluated at $n = 256$ and averaged over problems with equal weight. We compute the ratio as $\prod_{i=0}^{k-1} (n - c_q - i) / (n - i)$ to avoid large binomial coefficients. At $k = n$ the estimator reduces to an indicator of whether the pool contains any correct response, so PASS@256 equals the observed coverage of that pool.

Correctness. Correctness is always recomputed from the response text; we do not reuse the reward labels stored during rollout, which are produced by a different scorer and under a different generation setting. All three benchmarks have non-negative integer answers, and the verifier extracts a predicted integer with the first rule that applies: the content of the last `\boxed{...}` holding an integer; otherwise the last integer introduced by an explicit answer cue such as `final answer is`; otherwise the response as a whole, if it consists of a single integer. A response with no extractable integer counts as incorrect. Prediction and reference are compared after integer normalization, so the verifier accepts no symbolic equivalence beyond leading-zero and whitespace differences.

The second and third rules make the verifier permissive: a response can be scored correct without ever emitting a boxed answer. This matters for the coverage audit, where accepting an accidental hit from S_{INIT} would weaken the claim it supports. We therefore inspect every S_{INIT} recovery found in the extended pools by hand rather than trusting the verifier alone, and report the outcome per problem in Appendix B.

A.4 Prompt Templates

Training and evaluation examples contain one user message and no system message. Each training example already includes the instruction `Please reason step by step, and put your final answer within \boxed{...}`. after the problem statement, separated by a newline in DeepMath and a space in OpenThoughts3. We include the same instruction for mathematics evaluation and serialize the message with the student’s chat template. Routine evaluation uses the same template as training. Below, {QUESTION} denotes the problem statement without this instruction.

JustRL and DeepScaleR settings. Both use the DeepSeek-R1-Distill-Qwen-1.5B student template:

JustRL / DeepScaleR Prompt | DeepSeek-R1-Distill-Qwen-1.5B

```
<|begin_of_sentence|><|User|>{QUESTION}
Please reason step by step, and put your final answer within \boxed{.}<|Assistant|><think>
```

The assistant prefix ends with `<think>` followed by a newline, leaving the reasoning block open for generation.

Qwen3-4B setting. We use the Qwen3-1.7B student template with thinking mode disabled. Specifically, we prefill the assistant turn with an empty reasoning block, `<think>` followed by two newlines, `</think>`, and two more newlines. Generation therefore starts after the closed block. The serialized prompt is:

Qwen3-4B Setting Prompt | Qwen3-1.7B | Thinking Disabled

```
<|im_start|>user
{QUESTION} Please reason step by step, and put your final answer within \boxed{.}<|im_end|>
<|im_start|>assistant
<think>

</think>
```

Disabling thinking mode changes this assistant prefix; the user instruction to reason step by step remains in the prompt.

Teacher scoring. After the student generates a response, we feed the full token sequence—the serialized prompt and the student-generated response—into the teacher in a single forward pass to obtain the next-token distributions at the response positions for distillation. The teacher generates no new tokens (`max_new_tokens=0`). We use the student’s original token sequence directly, without reapplying a chat template or adding any instructions.

B Observed Solvable Sets

We first construct candidate solvable sets from problems with at least one verifier-correct response in the original 256-response pools. We then refine these sets through targeted sampling and manual review: invalid derivations are excluded, and valid solutions missed by the automatic verifier are reinstated. Table 6 reports the resulting audited partition of all 90 AIME24–26 problems under each teacher. The four columns form a disjoint partition: covered by both S_{INIT} and S_{OPD} , covered only by one of them, or covered by neither. Problem numbers are listed explicitly within each dataset.

The audit combines additional sampling with review of the existing outputs. We take every problem that is covered by S_{OPD} but not by S_{INIT} among 256 responses and extend the S_{INIT} pool to 1,024 responses. This recovers seven cases: JustRL AIME25-025 and AIME26-010, and DeepScaleR AIME25-014, AIME25-023, AIME25-025, AIME26-010, and AIME26-022. Manual review also checks the validity of S_{OPD} derivations among the candidate problems, including cases already recovered by additional sampling. The JustRL AIME25-020 and DeepScaleR AIME25-014 and AIME25-025 derivations are invalid despite reaching the correct final answer. The latter two problems are therefore assigned to the S_{INIT} -only set.

For JustRL AIME26-022 and AIME26-027, the automatic verifier records no S_{INIT} hit in the 1,024-response pools. Manual review identifies valid S_{INIT} solutions in the existing outputs and corrects these verifier misses; it does not use a separate batch of generations. These cases are assigned to the shared set. Table 7 distinguishes recoveries by additional sampling from corrections to automatic verification.

Table 6: **Audited observed-solvable-set partitions.** Each cell gives the number of problems followed by their three-digit suffixes.

Teacher	Data	Both	S_{INIT} only	S_{OPD} only	Neither
JustRL	AIME24	24 : 001–007, 009, 010, 013–022, 025–029	1 : 023	0 : –	5 : 008, 011, 012, 024, 030
JustRL	AIME25	18 : 001–006, 008, 009, 012, 016, 017, 019, 021, 022, 024, 025, 027, 029	6 : 010, 013, 014, 023, 028, 030	0 : –	6 : 007, 011, 015, 018, 020, 026
JustRL	AIME26	23 : 001–008, 010, 012–014, 016, 018–027	1 : 011	0 : –	6 : 009, 015, 017, 028–030
DeepScaleR	AIME24	23 : 001–007, 009, 010, 013–022, 025–028	2 : 023, 029	0 : –	5 : 008, 011, 012, 024, 030
DeepScaleR	AIME25	18 : 001–006, 008, 009, 012, 016, 017, 019, 021–024, 027, 029	6 : 010, 013, 014, 025, 028, 030	0 : –	6 : 007, 011, 015, 018, 020, 026
DeepScaleR	AIME26	22 : 001–008, 010, 012–014, 016, 018–026	2 : 011, 027	0 : –	6 : 009, 015, 017, 028–030

Table 7: **Resolution of the raw S_{OPD} -only cases.** The third column records how a correct S_{INIT} response was obtained. “Extended sampling” means the verifier accepted a response drawn while extending the pool to 1,024. “Manual review” means a correct response was already present and had been missed by the verifier, so no additional generation was needed. “None” means neither route produced one.

Teacher	Problem	S_{INIT} solution found by	S_{OPD} derivation	Resolution
JustRL	AIME25-020	None	Invalid	Removed as an accidental verifier hit
JustRL	AIME25-025	Extended sampling	Valid	Shared
JustRL	AIME26-010	Extended sampling	Valid	Shared
JustRL	AIME26-022	Manual review	Valid	Shared
JustRL	AIME26-027	Manual review	Valid	Shared
DeepScaleR	AIME25-014	Extended sampling	Invalid	S_{INIT} only
DeepScaleR	AIME25-023	Extended sampling	Valid	Shared
DeepScaleR	AIME25-025	Extended sampling	Invalid	S_{INIT} only
DeepScaleR	AIME26-010	Extended sampling	Not needed	Shared
DeepScaleR	AIME26-022	Extended sampling	Valid (two re-views)	Shared

These sets describe the responses observed under the stated sampling and manual-audit protocol. They do not establish equality of model support or zero probability for any response under S_{INIT} .

C Min-10NN Distance

Min-10NN measures how tightly responses to the same problem concentrate in surface form. For two decoded responses a and b , we use character-level normalized Levenshtein distance

$$d(a, b) = \begin{cases} 0, & |a| = |b| = 0, \\ \text{Lev}(a, b) / \max\{|a|, |b|\}, & \text{otherwise.} \end{cases} \quad (6)$$

Let $R = \{r_i\}_{i=1}^{16}$ be 16 responses to one problem. For each response, we compute its distance to the nearest other response,

$$\delta_i = \min_{j \neq i} d(r_i, r_j). \quad (7)$$

After sorting these values as $\delta_{(1)} \leq \dots \leq \delta_{(16)}$, Min-10NN is

$$\text{Min-10NN}(R) = \frac{1}{10} \sum_{i=1}^{10} \delta_{(i)}. \quad (8)$$

A smaller value means that the ten most tightly clustered responses are more similar. When 32 responses are available, we sample 16 without replacement, average Min-10NN over 100 repetitions, and then average across problems. The metric captures character-level concentration only; it does not measure semantic equivalence or response quality.

D Response Case Study

Figure 10 gives the examples of the excerpts condensed in Figure 1b. All four responses come from the Qwen3-4B setting before any OPD update: (a) is a free generation of the teacher, and (b)–(d) are sampled from S_{INIT} , with (c) and (d) drawn from the responses the teacher ranks as most preferred under the decile partition of Section 5.1.

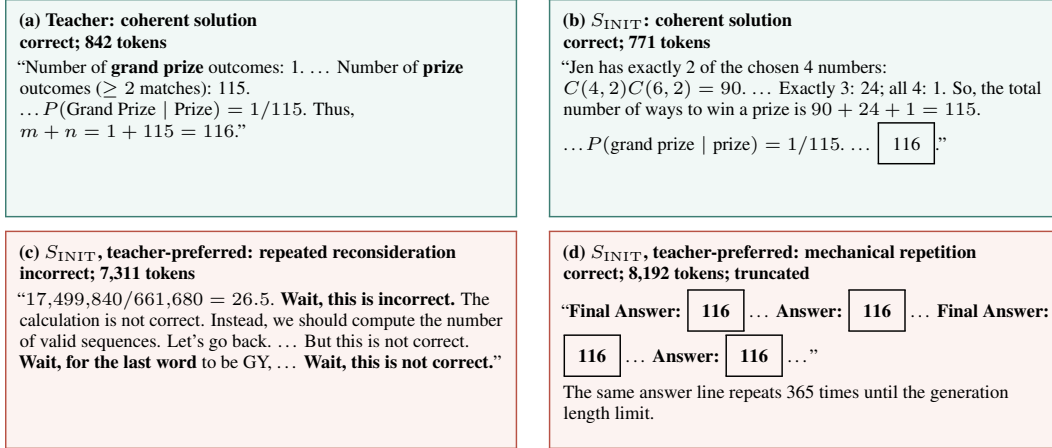


Figure 10: **Illustrative response excerpts from the Qwen3-4B setting.** The teacher and a typical S_{INIT} response give coherent solutions, whereas the S_{INIT} responses the teacher most prefers repeatedly restart their reasoning or mechanically repeat an answer until the generation length limit. All four excerpts predate any OPD update. Ellipses mark omitted text. These examples illustrate behaviors and do not estimate their prevalence.

E High-Advantage Responses from the Initial Student

We present three responses sampled from Qwen3-1.7B-Base before any OPD update and scored by the Qwen3-4B teacher. All three belong to the highest-advantage group, A10, in the analysis of Section 5.1 and illustrate excessive length and mechanical repetition already present under S_{INIT} . All three are incorrect and contain no parseable final answer.

Reading the complete responses. Each response consists almost entirely of a repeated string. We specify its complete content using exact repetition counts: $+$ denotes concatenation without an added separator, and $\text{repeat}(s, n)$ denotes n consecutive copies of s . Quotation marks delimit strings and are not part of the response; $\backslash n$ denotes a newline. Unicode characters are written as $\backslash uXXXX$ escapes. No text is omitted from these representations. The original UTF-8 prompts, responses, token IDs, and provenance are retained in the accompanying data/appendix/sinit_examples/ directory. Prompt mathematics below is typeset for readability. Every prompt ends with: “Please reason step by step, and put your final answer within $\boxed{}$.”

E.1 Length-Cap Repetition

Prompt (AIME24, aime24_001). Every morning Aya goes for a 9-kilometer-long walk and stops at a coffee shop afterwards. When she walks at a constant speed of s kilometers per hour, the walk takes her 4 hours, including t minutes spent in the coffee shop. When she walks $s + 2$ kilometers per hour, the walk takes her 2 hours and 24 minutes, including t minutes spent in the coffee shop. Suppose Aya walks at $s + \frac{1}{2}$ kilometers per hour. Find the number of minutes the walk takes her, including the t minutes spent in the coffee shop.

Response (sample 65).

```
"\uFFFD\uFFFD\u6001 T 10T 50 10 1 10 10"
+ repeat("T 10", 2043)
```

The response mechanically repeats T 10 until it reaches the 8,192-token generation cap. The two initial $\backslash uFFFD$ characters are present in the stored response.

E.2 Severe Repetition without Reaching the Length Cap

Prompt (AIME25, aime25_028). Let the sequence of rationals $x_1, x_2, \dots$ be defined such that $x_1 = \frac{25}{11}$ and $x_{k+1} = \frac{1}{3} \left(x_k + \frac{1}{x_k} - 1 \right)$. x_{2025} can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n . Find the remainder when $m + n$ is divided by 1000.

Response (sample 110).

```
repeat(".slfwith\n", 2154)+ ".slfwith"
```

The response consists of repeated .slfwith lines and ends after 6,465 recorded tokens, showing that severe repetition can occur without reaching the generation cap.

E.3 Severe Repetition at the Length Cap

Prompt (AIME25, aime25_013). Alex divides a disk into four quadrants with two perpendicular diameters intersecting at the center of the disk. He draws 25 more line segments through the disk, drawing each segment by selecting two points at random on the perimeter of the disk in different quadrants and connecting those two points. Find the expected number of regions into which these 27 line segments divide the disk.

Response (sample 58).

```
"\u6CD5 newVa189\u6CD5 newVa18\u6CD5 newVa18"
+ repeat("\u6CD5 newVa1", 4091)
```

The response repeats the same fragment until it reaches the 8,192-token generation cap, exhibiting both excessive length and severe repetition.

Together, these cases illustrate how excessively long and repetitive responses can receive high relative teacher advantage already under S_{INIT} .

F Training Details for the Candidate-Pool Interventions

Section 5.2 compares three Qwen3-4B teacher settings. The baseline and masked settings initialize the student from Qwen3-1.7B-Base. The warmup setting instead initializes it from the Qwen3-1.7B-SFT checkpoint released by Li et al. [2026] and uses that model as the fixed reference. All three use OpenThoughts3 prompts, temperature 1.0, an 8,192-token generation length limit, and the optimization settings in Appendix A.1.

Loss masking for truncated or failed responses. The baseline trains on all sampled responses, including those that reach the generation length limit. In the masked setting, we set the loss mask to zero for truncated or failed responses; completed responses retain nonzero loss masks and contribute to training. Truncated or failed responses remain in reward and advantage normalization but contribute no policy, reference-KL, or entropy gradient. The loss is first averaged over unmasked response tokens within each sample and then over samples in the global batch.

Initialization from a released SFT checkpoint. The warmup setting directly loads the Qwen3-1.7B-SFT checkpoint released by Li et al. [2026]; we do not perform an additional SFT stage ourselves. It changes only the student initialization and fixed reference from Qwen3-1.7B-Base to Qwen3-1.7B-SFT; the teacher, data, optimizer, sampling protocol, and evaluation schedule remain fixed. Its higher initial accuracy is part of the intervention: it tests whether supplying a better initial candidate pool changes what OPD can amplify.